

Problem Set #28

UMass Recreational Math Club

February 2026

Problem 1. Put these numbers in order: 1999^{1999} , 2000^{1998} , 1998^{2000} .

Problem 2. Somebody draws 9 straight lines on the plane and finds out that each of them cuts a fixed square into two quadrilateral pieces with areas in ratio 2 : 1. Prove that at least 3 of these lines have a common point.

Problem 3. Let p be a prime number. Prove that $(p-1)! \equiv -1 \pmod{p}$.

Bonus: can you prove that this is *only* true if p is prime?

Problem 4. Carol and Desmond are playing pool with billiard balls number 1 through 9. They take turns sinking balls into pockets. The first to sink three numbers that sum to 15 wins. Does Carol (the first to play) have a winning strategy?